| Database Systems | |
|---|---|
| **Lab #:** | **09** |
| **Lab Title:** | Implementation of DML commands of SQL with suitable examples |
| **Submitted by:** | |

| **Names** | **Registration #** |
|---|---|
| Sumaira Safeer | FA22-BCE-019 |

| **Rubrics name & number** | | **Marks** | |
|---|---|---|---|
| | | **In-Lab** | **Post-Lab** |
| **Engineering Knowledge** | ***R2: Use of Engineering Knowledge and follow Experiment Procedures:*** *Ability to follow experimental procedures, control variables, and record procedural steps on lab report.* | | |
| **Problem Analysis** | ***R6: Experimental Data Analysis:*** *Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.* | | |
| **Design** | ***R8: Best Coding Standards:*** *Ability to follow the coding standards and programming practices.* | | |
| **Modern Tools Usage** | ***R9: Understand Tools:*** *Ability to describe and explain the principles behind and applicability of engineering tools.* | | |
| **Individual and Teamwork** | ***R12: Individual Work Contributions:*** *Ability to carry out individual responsibilities.* | | |
| | ***R13: Management of Team Work:*** *Ability to appreciate, understand and work with multidisciplinary team members.* | | |

| **Rubrics #** | **R2** | **R6** | **R8** | **R9** | **R12** | **R13** |
|---|---|---|---|---|---|---|
| **In Lab** | | | | | | |
| **Post- Lab** | | | | | | |

# Lab No: 09

## Title: Implementation of DML commands of SQL with suitable examples.

- Insert table.
- Update table.
- Delete Table.

Implementation of different types of functions with suitable examples:

- Number Function.
- Aggregate Function.

## Objective:

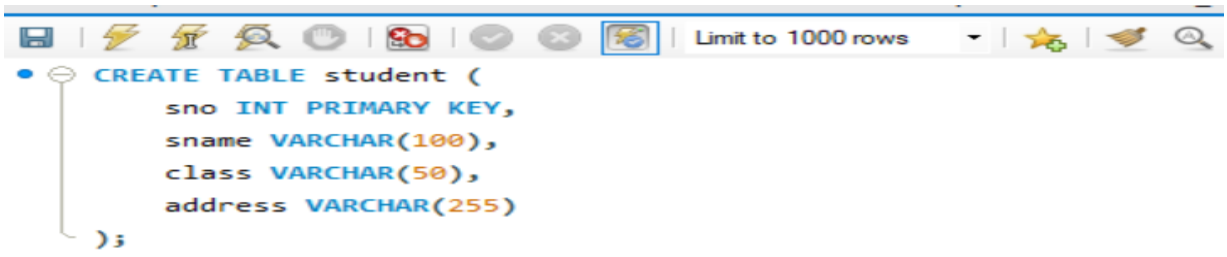
To understand the different issues involved in the design and implementation of a database system.

To understand and use data manipulation language to query, update, and manage a database.

To understand and implement various types of function in SQL.

## InLab Tasks:

**Inserting the Student Records:**



```sql
select * from student;
INSERT INTO student (sno, sname, class, address) VALUES
(1, 'Sumaira Safeer', '10th Grade', 'Karachi, Sindh, Pakistan'),
(2, 'Rohan  Khan', '12th Grade', 'Lahore, Punjab, Pakistan'),
(3, 'Ali Mustafa', '9th Grade', 'Islamabad, Capital Territory, Pakistan'),
(4, 'Nayab Bibi', '10th Grade', 'Rawalpindi, Punjab, Pakistan'),
(5, 'Kamran Hassan', '11th Grade', 'Multan, Punjab, Pakistan'),
(6, 'Yasir Malik', '12th Grade', 'Faisalabad, Punjab, Pakistan'),
(7, 'Maria Farooq', '8th Grade', 'Peshawar, Khyber Pakhtunkhwa, Pakistan'),
(8, 'Mariam Nasir', '9th Grade', 'Quetta, Balochistan, Pakistan'),
(9, 'Hashim Ahmed', '11th Grade', 'Karachi, Sindh, Pakistan'),
(10, 'Amna Iqbal', '10th Grade', 'Sialkot, Punjab, Pakistan');
```

| | | | | | |
|---|---|---|---|---|---|
| ✓ | 1 | 14:16:25 | CREATE TABLE student (  sno INT PRIMARY KEY,  sname VARCHAR(100)... | 0 row(s) affected | 0.031 sec |
| ✓ | 2 | 14:16:25 | select * from student LIMIT 0, 1000 | 0 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ | 3 | 14:16:25 | INSERT INTO student (sno, sname, class, address) VALUES (1, 'Ahmed Ali', '10t... | 10 row(s) affected Records: 10  Duplicates: 0  Warnings: 0 | 0.016 sec |

**Inserting the  product Records:**

```sql
CREATE TABLE product (
    products_id INT PRIMARY KEY,
    product_name VARCHAR(50),
    pro_code INT,
    price DECIMAL(10, 2)
);

INSERT INTO product (products_id, product_name, pro_code, price) VALUES
(1, 'T_shirt', 768, 7800),
(2, 'shirt', 124, 599),
(3, 'lipstick', 948, 800),
(4, 'lense', 7976, 1500),
(5, 'jeans', 234, 1200),
(6, 'jacket', 451, 3000),
(7, 'shoes', 867, 4500),
(8, 'hat', 342, 750),
(9, 'belt', 556, 999),
(10, 'sunglasses', 221, 2000);
```

## Task1:

- Update the Shirt with shoes by applying condition on Pro_code in product detail table
- Update the T_Shirt with Shirt and 599 with 600 by applying condition on Pro_code.

**Syntax:** SQL>UPDATE relation name SET Field_name1=data,field_name2=data,

WHERE field_name=data;

```
1  INSERT INTO products(products_id,product_name,pro_code,price)VALUES (1,'T_shirt','768',7800);
2  INSERT INTO products(products_id,product_name,pro_code,price)VALUES (2,'Shoes','124',600);
3  INSERT INTO products(products_id,product_name,pro_code,price)VALUES (3,'lipstick','0948',800);
4  INSERT INTO products(products_id,product_name,pro_code,price)VALUES (4,'lense','7976',1500);
5
6  UPDATE products SET product_name = 'shirt' WHERE products_id=2;
7  UPDATE products SET price = '599' WHERE products_id=2;
8
```

## Result Grid:

Result Grid | Filter Rows:

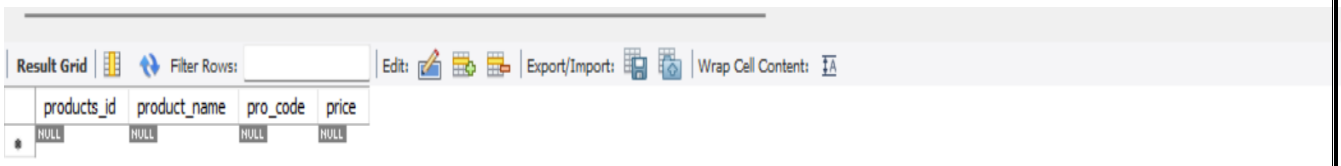
| products_id | product_name | pro_code | price |
|---|---|---|---|
| 1 | T_shirt | 768 | 7800 |
| 2 | shirt | 124 | 599 |
| 3 | lipstick | 948 | 800 |
| 4 | lense | 7976 | 1500 |
| NULL | NULL | NULL | NULL |

## Task 2:

- Delete all records from Product-detail.

**Deleting a record:**

```
39
40 •    select * from product;
41 •    DELETE FROM product;
42
```



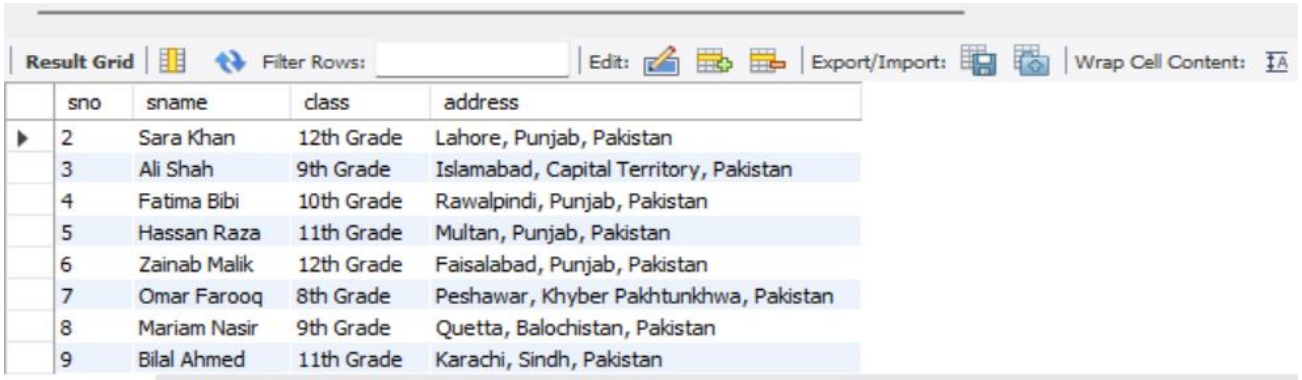
| products_id | product_name | pro_code | price |
|---|---|---|---|
| NULL | NULL | NULL | NULL |

| | | | | | |
|---|---|---|---|---|---|
| 13 | 22:25:11 | SET SQL_SAFE_UPDATES = 0 | 0 row(s) affected | 0.000 sec |
| 14 | 22:25:11 | DELETE FROM product | 10 row(s) affected | 0.015 sec |
| 15 | 22:25:11 | SET SQL_SAFE_UPDATES = 1 | 0 row(s) affected | 0.000 sec |
| 16 | 22:25:17 | select * from product LIMIT 0, 1000 | 0 row(s) returned | 0.000 sec / 0.000 sec |

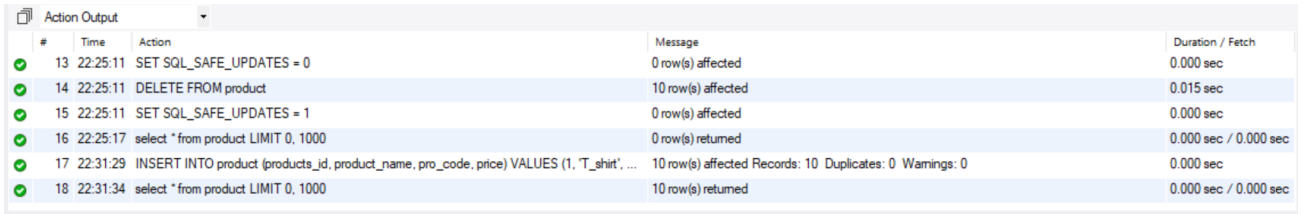
• Delete a record from Student_info where St_id=1

**Deleting a record:**

```
38
39 •    DELETE FROM student WHERE sno = 1;
40
```



| sno | sname | class | address |
|---|---|---|---|
| 2 | Sara Khan | 12th Grade | Lahore, Punjab, Pakistan |
| 3 | Ali Shah | 9th Grade | Islamabad, Capital Territory, Pakistan |
| 4 | Fatima Bibi | 10th Grade | Rawalpindi, Punjab, Pakistan |
| 5 | Hassan Raza | 11th Grade | Multan, Punjab, Pakistan |
| 6 | Zainab Malik | 12th Grade | Faisalabad, Punjab, Pakistan |
| 7 | Omar Farooq | 8th Grade | Peshawar, Khyber Pakhtunkhwa, Pakistan |
| 8 | Mariam Nasir | 9th Grade | Quetta, Balochistan, Pakistan |
| 9 | Bilal Ahmed | 11th Grade | Karachi, Sindh, Pakistan |



| # | Time | Action | Message | Duration / Fetch |
|---|---|---|---|---|
| 13 | 22:25:11 | SET SQL_SAFE_UPDATES = 0 | 0 row(s) affected | 0.000 sec |
| 14 | 22:25:11 | DELETE FROM product | 10 row(s) affected | 0.015 sec |
| 15 | 22:25:11 | SET SQL_SAFE_UPDATES = 1 | 0 row(s) affected | 0.000 sec |
| 16 | 22:25:17 | select * from product LIMIT 0, 1000 | 0 row(s) returned | 0.000 sec / 0.000 sec |
| 17 | 22:31:29 | INSERT INTO product (products_id, product_name, pro_code, price) VALUES (1, 'T_shirt', ... | 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 | 0.000 sec |
| 18 | 22:31:34 | select * from product LIMIT 0, 1000 | 10 row(s) returned | 0.000 sec / 0.000 sec |

## Task 3:

- Display all records of Product_detail and Student_info.

  **Syntax:** SELECT * FROM relation_name;

**Display of Student_info:**

```
 7 •   select * from student;
 8 •   INSERT INTO student (sno, sname, class, address) VALUES
 9     (1, 'Sumaira Safeer', '10th Grade', 'Karachi, Sindh, Pakistan'),
10     (2, 'Rohan  Khan', '12th Grade', 'Lahore, Punjab, Pakistan'),
11     (3, 'Ali Mustafa', '9th Grade', 'Islamabad, Capital Territory, Pakistan'),
12     (4, 'Nayab Bibi', '10th Grade', 'Rawalpindi, Punjab, Pakistan'),
13     (5, 'Kamran Hassan', '11th Grade', 'Multan, Punjab, Pakistan'),
14     (6, 'Yasir Malik', '12th Grade', 'Faisalabad, Punjab, Pakistan'),
15     (7, 'Maria Farooq', '8th Grade', 'Peshawar, Khyber Pakhtunkhwa, Pakistan'),
16     (8, 'Mariam Nasir', '9th Grade', 'Quetta, Balochistan, Pakistan'),
17     (9, 'Hashim Ahmed', '11th Grade', 'Karachi, Sindh, Pakistan'),
18     (10, 'Amna Iqbal', '10th Grade', 'Sialkot, Punjab, Pakistan');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| sno | sname | class | address |
|---|---|---|---|
| 1 | Ahmed Ali | 10th Grade | Karachi, Sindh, Pakistan |
| 2 | Sara Khan | 12th Grade | Lahore, Punjab, Pakistan |
| 3 | Ali Shah | 9th Grade | Islamabad, Capital Territory, Pakistan |
| 4 | Fatima Bibi | 10th Grade | Rawalpindi, Punjab, Pakistan |
| 5 | Hassan Raza | 11th Grade | Multan, Punjab, Pakistan |
| 6 | Zainab Malik | 12th Grade | Faisalabad, Punjab, Pakistan |
| 7 | Omar Farooq | 8th Grade | Peshawar, Khyber Pakhtunkhwa, Pakistan |
| 8 | Mariam Nasir | 9th Grade | Quetta, Balochistan, Pakistan |

**Display of Product_detail:**

```
35      (6, 'jacket', 451, 3000),
36      (7, 'shoes', 867, 4500),
37      (8, 'hat', 342, 750),
38      (9, 'belt', 556, 999),
39      (10, 'sunglasses', 221, 2000);
40 •    select * from product;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

| products_id | product_name | pro_code | price |
|---|---|---|---|
| 1 | T_shirt | 768 | 7800.00 |
| 2 | shirt | 124 | 599.00 |
| 3 | lipstick | 948 | 800.00 |
| 4 | lense | 7976 | 1500.00 |
| 5 | jeans | 234 | 1200.00 |
| 6 | jacket | 451 | 3000.00 |
| 7 | shoes | 867 | 4500.00 |
| 8 | hat | 342 | 750.00 |

| | | | | |
|---|---|---|---|---|
| 15 | 22:25:11 | SET SQL_SAFE_UPDATES = 1 | 0 row(s) affected | 0.000 sec |
| 16 | 22:25:17 | select * from product LIMIT 0, 1000 | 0 row(s) returned | 0.000 sec / 0.000 sec |
| 17 | 22:31:29 | INSERT INTO product (products_id, product_name, pro_code, price) VALUES (1, 'T_shirt', ... | 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 | 0.000 sec |
| 18 | 22:31:34 | select * from product LIMIT 0, 1000 | 10 row(s) returned | 0.000 sec / 0.000 sec |

## **Task 4**:

- Display St_name and address form Student_info.

  **Syntax:** SELECT a set of fields FROM relation_name;

```
43
44 •    select sname, address from student;
45
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| sname | address |
|---|---|
| Sara Khan | Lahore, Punjab, Pakistan |
| Ali Shah | Islamabad, Capital Territory, Pakistan |
| Fatima Bibi | Rawalpindi, Punjab, Pakistan |
| Hassan Raza | Multan, Punjab, Pakistan |
| Zainab Malik | Faisalabad, Punjab, Pakistan |
| Omar Farooq | Peshawar, Khyber Pakhtunkhwa, Pakistan |
| Mariam Nasir | Quetta, Balochistan, Pakistan |
| Bilal Ahmed | Karachi, Sindh, Pakistan |

student 10 ×

Output

Action Output

| # | Time | Action | Message | Duration / Fetch |
|---|---|---|---|---|
| 13 | 22:25:11 | SET SQL_SAFE_UPDATES = 0 | 0 row(s) affected | 0.000 sec |

## Task 5:

- Display the record of each Student whose class is 9th Grade.

  **Syntax:** SELECT a set of fields FROM relation_name WHERE condition;

```
44
45 •    SELECT sno, sname, class, address FROM student WHERE class = '9th Grade';
46
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| sno | sname | class | address |
|---|---|---|---|
| 3 | Ali Shah | 9th Grade | Islamabad, Capital Territory, Pakistan |
| 8 | Mariam Nasir | 9th Grade | Quetta, Balochistan, Pakistan |
| NULL | NULL | NULL | NULL |

Action Output

| # | Time | Action | Message | Duration / Fetch |
|---|---|---|---|---|
| 21 | 22:40:40 | select * from student LIMIT 0, 1000 | 9 row(s) returned | 0.000 sec / 0.000 sec |
| 22 | 22:41:48 | select sname, address from student LIMIT 0, 1000 | 9 row(s) returned | 0.000 sec / 0.000 sec |
| 23 | 22:42:46 | select * from student LIMIT 0, 1000 | 9 row(s) returned | 0.000 sec / 0.000 sec |
| 24 | 22:45:40 | SELECT sname FROM student WHERE class = '9th Grade' LIMIT 0, 1000 | 2 row(s) returned | 0.000 sec / 0.000 sec |
| 25 | 22:45:59 | SELECT sname, class FROM student WHERE class = '9th Grade' LIMIT 0, 1000 | 2 row(s) returned | 0.000 sec / 0.000 sec |
| 26 | 22:49:00 | SELECT sno, sname, class, address FROM student WHERE class = '9th Grade' LIMIT 0, 10... | 2 row(s) returned | 0.016 sec / 0.000 sec |

## Task 6:

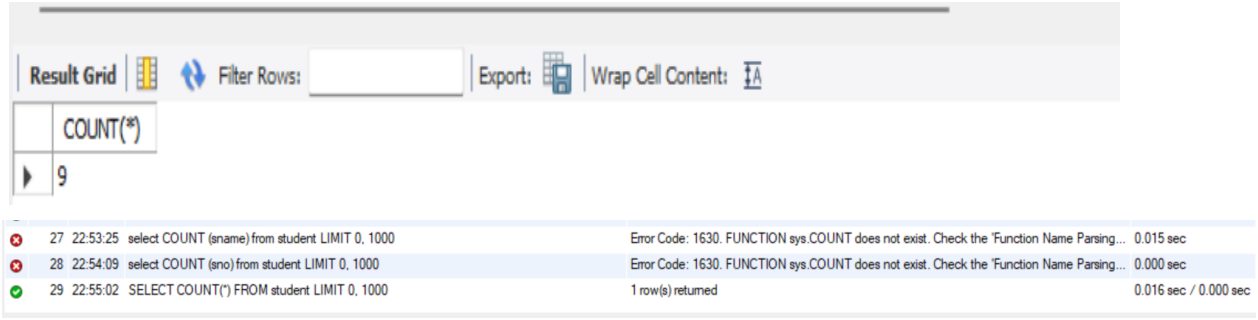
- Count the number of students in Student_info.

  **Count:** COUNT following by a column name returns the count of tuple in that column.

  **Syntax:** COUNT (Column name).

  **Example:** SELECT COUNT (Sal) FROM emp;

```
46
47 •    SELECT COUNT(*) FROM student;
48
```




## Task 7:

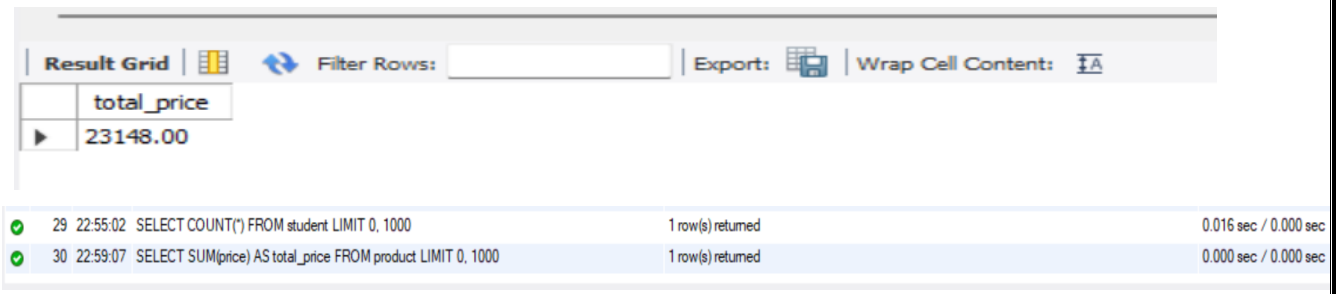
- Add all the product price form the Product detail table.

  **SUM:** SUM followed by a column name returns the sum of all the values in that column.

  **Syntax:** SUM (Column name).

  **Example:** SELECT SUM (Sal) From emp;

```
47
48 •    SELECT SUM(price) AS total_price FROM product;
49
```




## Task 8:

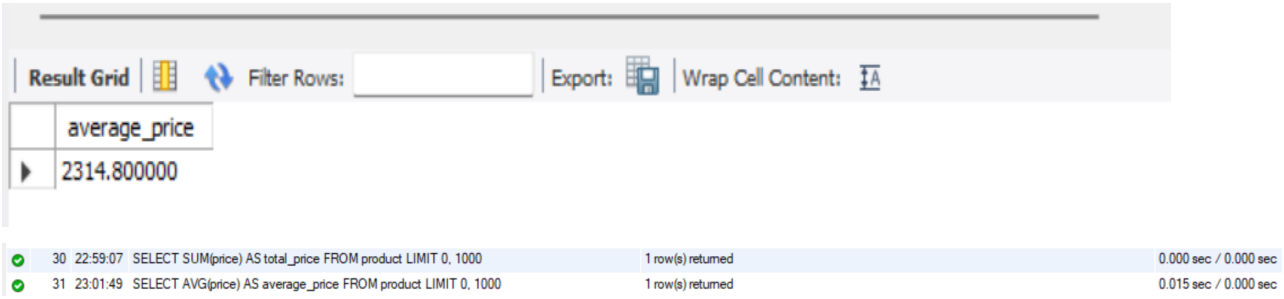
- Find the average age from the price_info table.

  **AVG:** AVG followed by a column name returns the average value of that column values**.**

  **Syntax:** AVG (n1, n2...)

  **Example:** Select AVG (10, 15, 30) FROM DUAL;

```
48
49 •    SELECT AVG(price) AS average_price FROM product;
50
```




## Task 9:

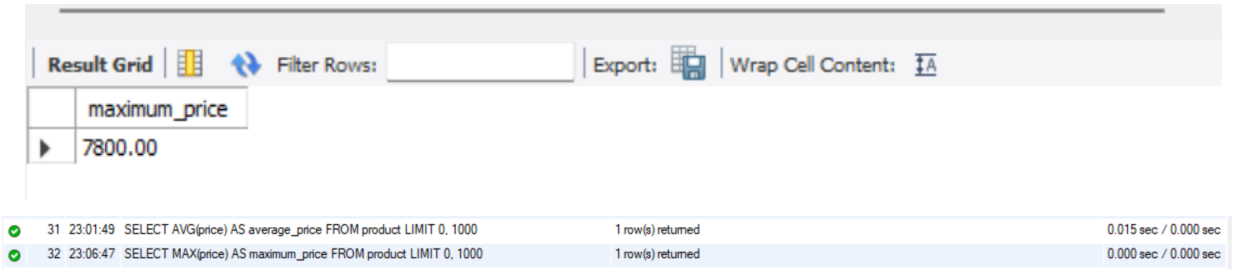
- Find the maximum Pro_Price in the Product_detail table.

  **MAX:** MAX followed by a column name returns the maximum value of that column.

  **Syntax:** MAX (Column name).

  **Example:** SELECT MAX (Sal) FROM emp;

```
49
50 •    SELECT MAX(price) AS maximum_price FROM product;
51
```




## Task 10:

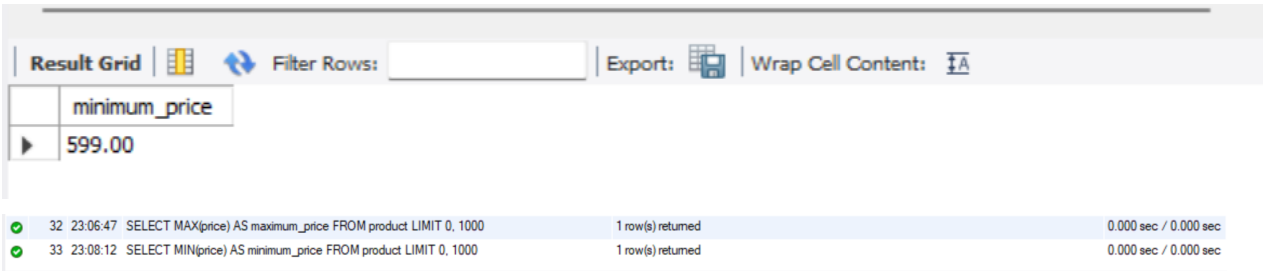
- Find the minimum Pro_Price in the Product_detail table.

  **MIN:** MIN followed by column name returns the minimum value of that column.

  **Syntax:** MIN (Column name).

**Example:** SELECT MIN (Sal) FROM emp;

```
51
52 •    SELECT MIN(price) AS minimum_price FROM product;
53      ----------------------------------------------------------------------------
54
```




## Post lab task:

### Task 1:

Write a difference between delete and truncate command.

**Answer:**

**TRUNCATE:**

- TRUNCATE is a DDL command
- TRUNCATE is executed using a table lock and whole table is locked for remove allrecords.
- We cannot use Where clause with TRUNCATE.
- TRUNCATE removes all rows from a table.
- Minimal logging in transaction log, so it is performance wise faster.
- TRUNCATE TABLE removes the data by deallocating the data pages used to store thetable data and records only the page deallocations in the transaction log.
- Identify column is reset to its seed value if table contains any identity column.
- To use Truncate on a table you need at least ALTER permission on the table.
- Truncate uses the less transaction space than Delete statement.
- Truncate cannot be used with indexed views.

**DELETE:**

- DELETE is a DML command.
- DELETE is executed using a row lock, each row in the table is locked for deletion.
- We can use where clause with DELETE to filter & delete specific records.
- The DELETE command is used to remove rows from a table based on WHEREcondition.
- It maintain the log, so it slower than TRUNCATE.
- The DELETE statement removes rows one at a time and records an entry in thetransaction log for each deleted
- Identity of column keep DELETE retain the identity.
- To use Delete you need DELETE permission on the table.
- Delete uses the more transaction space than Truncate statement.
- Delete can be used with indexed views.

## Task2:

Create a table EMPLOYEE with following schema:

(Emp_no, E_name, E_address, E_ph_no, Dept_no, Dept_name,Job_id, Designation , Salary)

## Query:

```
create database employee;
use employee;
create table employee(Emp_no int(30), E_name varchar(30), E_address varchar(40), E_pho_no varchar(20), Dept_no varchar(20),
Dept_name varchar(30), job_id varchar(20), Designation varchar(20), salary int(6));

insert into employee values(1, 'Muhammad Arsalan', 'Arsalankhan26666@gmail.com', '0300-3976820', '1', 'ECE', '025',
'Student', 20000);

insert into employee values(1, 'Micheal Jackson', 'Jackson.micheal1@gmail.com', '0300-21158657', '2', 'Cs', '235',
'Dean', 270000);

insert into employee values(1, 'Nauman Arshad', 'ANauman.arshad@outlook.com', '0344-39478564', '3', 'Management', '431',
'Lecturer', 110000);

insert into employee values(1, 'Hamid Ali', 'Hamidkhan@hotmail.com', '0315-1698741', '4', 'Mathematics', '511',
'Lab Engineer', 45000);

insert into employee values(1, 'Muhammad Adil', 'adiladil@hotmail.com', '0314-1290820', '5', 'Sweeping', '740',
'Sweeper', 15500);
```

Filter: Export: Autosize:

| Emp_no | E_name | E_address | E_pho_no | Dept_no | Dept_name | job_id | Designation | salary |
|---|---|---|---|---|---|---|---|---|
| 1 | Muhammad Arsalan | Arsalankhan26666@gmail.com | 0300-3976820 | 1 | ECE | 025 | Student | 20000 |
| 1 | Micheal Jackson | Jackson.micheal1@gmail.com | 0300-21158657 | 2 | Cs | 235 | Dean | 270000 |
| 1 | Nauman Arshad | ANauman.arshad@outlook.com | 0344-39478564 | 3 | Management | 431 | Lecturer | 110000 |
| 1 | Hamid Ali | Hamidkhan@hotmail.com | 0315-1698741 | 4 | Mathematics | 511 | Lab Engineer | 45000 |
| 1 | Muhammad Adil | adiladil@hotmail.com | 0314-1290820 | 5 | Sweeping | 740 | Sweeper | 15500 |

**Write SQL statements for the following query.**

1. List the E_no, E_name, Salary of all employees working for MANAGER.

```
select Emp_no, E_name, salary from employee where Designation='Manager';
```

Filter: Export: Autosize:

| Emp_no | E_name | salary |
|---|---|---|
| 1 | Muhammad Arsal... | 220000 |
| 1 | Micheal Jackson | 270000 |

2. Display all the details of the employee whose salary is more than the Sal of any IT

```
22
23 •  select * from employee where salary>110000;
```

Professor.

Filter: Export: Autosize:

| Emp_no | E_name | E_address | E_pho_no | Dept_no | Dept_name | job_id | Designation | salary |
|---|---|---|---|---|---|---|---|---|
| 1 | Muhammad Arsalan | Arsalankhan26666@gmail.com | 0300-3976820 | 1 | ECE | 025 | Manager | 220000 |
| 1 | Micheal Jackson | Jackson.micheal1@gmail.com | 0300-21158657 | 2 | Cs | 235 | Manager | 270000 |

3. List the employees who are 'CLERK'.

```
25 •  select * from employee where Designation='Clerk';
```

Filter: Export: Autosize:

| Emp_no | E_name | E_address | E_pho_no | Dept_no | Dept_name | job_id | Designation | salary |
|---|---|---|---|---|---|---|---|---|
| 1 | Muhammad Adil | adiladil@hotmail.com | 0314-1290820 | 5 | Sweeping | 740 | Clerk | 15500 |

**Conclusion:**

In this lab we implement different type of commands insert, update, delete. We also usein this lab where command for setting condition